

# C05 – ADVANCED REINFORCEMENT LEARNING ASSESSMENT

## Simulation-Based Assessment

### STUDENT DETAILS

|                            |                                 |                       |             |
|----------------------------|---------------------------------|-----------------------|-------------|
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| <b>Programme / Section</b> | Slot B                          | <b>Date</b>           | 19-08-2026  |
| <b>Course Code</b>         | C05                             | <b>Course Outcome</b> | C05         |
| <b>Assessment Tool</b>     | Simulation-Based Assessment     | <b>AT No.</b>         | 02          |
| <b>Assessment Type</b>     | Advanced Reinforcement Learning | <b>Bloom Level</b>    | L3 / L4     |
| <b>Faculty</b>             | Dr. Vincent Gnanaraj T          | <b>Marks</b>          | ____ / ____ |

### Assessment Details

#### C05 – Advanced Reinforcement Learning Assessment

#### Assessment Question – Hierarchical RL & MAXQ (Apply – L3)

#### Scenario-Based Assessment – Meta-Learning, Multi-Agent RL & POMDP (Analyse – L4)

Instruction: Each question contains TWO missing blocks. Select the option pair that correctly completes BOTH blanks.

## Assessment Question – Hierarchical RL & MAXQ (Apply – L3)

1. A warehouse robot uses a high-level policy to choose a task and a low-level policy to execute actions. HRL therefore separates \_\_\_\_ decisions from \_\_\_\_ decisions.

A. local / global

B. strategic / operational

C. random / statistical

D. reward / state

2. A high-level controller selects NavigateRoom, while a lower controller performs wheel movements. The first controller works at a \_\_\_\_ time scale and the second at a \_\_\_\_ time scale.

A. slower / faster

B. faster / slower

C. local / global

D. fixed / random

3. A robot learns NavigateToDoor once and reuses it for several missions. This demonstrates skill \_\_\_\_ and can improve learning \_\_\_\_.

A. duplication / latency

B. reuse / efficiency

C. encoding / noise

D. removal / variance

4. A long-horizon task has rewards only at the end. Splitting it into reusable subtasks can reduce the effective \_\_\_\_ horizon and improve \_\_\_\_.

A. reward / encryption

B. state / compression

C. policy / rendering

D. planning / exploration

5. A high-level policy chooses a sub-goal and waits until the skill finishes before selecting another. This uses temporal \_\_\_\_ and explicit skill \_\_\_\_.

A. randomization / encoding

B. abstraction / termination

C. normalization / sampling

D. compression / clipping

6. If an HRL controller changes sub-goals before a skill completes, the hierarchy may suffer from poor \_\_\_\_ design and unstable \_\_\_\_ control.

A. reward / visual

B. dataset / statistical

C. state / network

D. termination / temporal

7. A MAXQ hierarchy represents a mission using parent and child tasks. The main value function is therefore \_\_\_\_ into smaller task-level \_\_\_\_.

- A. normalized / episodes
- B. encrypted / states
- C. decomposed / values
- D. duplicated / datasets

8. In MAXQ, a child task may obtain a high local value but still reduce the parent task's outcome. Local optimization does not always imply global \_\_\_\_ and task-level \_\_\_\_.

- A. accuracy / encryption
- B. optimality / coordination
- C. variance / sampling
- D. randomness / compression

9. A parent task invokes Search, Pick, and Deliver as child tasks. This hierarchy supports task \_\_\_\_ and reusable \_\_\_\_.

- A. sampling / rewards
- B. compression / states
- C. encryption / gradients
- D. composition / subtasks

10. A designer encodes Scan → Identify → Pick → Place before learning. This reduces the exploration \_\_\_\_ but may reduce policy \_\_\_\_.

- A. space / flexibility
- B. dataset / variance
- C. reward / accuracy
- D. state / latency

11. A skill can be used by several parent tasks when its goal conditions are compatible. This property is called skill \_\_\_\_ and supports policy \_\_\_\_.

- A. reuse / modularity
- B. sampling / encryption
- C. duplication / latency
- D. clipping / variance

12. A subtask takes different numbers of primitive actions in different states. Its value must account for variable \_\_\_\_ and accumulated \_\_\_\_.

- A. width / memory
- B. depth / pixels
- C. frequency / noise
- D. duration / reward

13. An over-constrained hierarchy prevents the agent from discovering a shorter route. The structure improves guidance but reduces behavioral \_\_\_\_ and task \_\_\_\_.

- A. flexibility / adaptability
- B. variance / compression
- C. latency / accuracy
- D. reward / sampling

14. A high-level policy chooses Explore, and the lower policy decides movement. The high-level action is an abstract \_\_\_\_ while movement is a primitive \_\_\_\_.

- A. reward / state
- B. state / skill
- C. skill / action
- D. gradient / episode

15. A MAXQ parent combines child-task values with the effects of completing those children. This represents hierarchical \_\_\_\_ rather than a single flat \_\_\_\_.

- A. encryption / reward
- B. compression / episode
- C. randomization / dataset
- D. decomposition / policy

16. A meta-RL agent trains on many traffic patterns and adapts quickly to a new pattern. It learns a transferable \_\_\_\_ rather than one fixed \_\_\_\_.

- A. gradient / episode
- B. dataset / reward
- C. state / action
- D. strategy / policy

17. A meta-trained agent adapts poorly to unrelated tasks. This indicates that meta-learning depends on the \_\_\_\_ between tasks and may fail under distribution \_\_\_\_.

- A. relatedness / shift
- B. reward / clipping
- C. policy / compression
- D. state / encoding

18. A meta-RL system learns an initialization that changes rapidly using few new samples. This improves adaptation \_\_\_\_ and reduces data \_\_\_\_.

- A. speed / requirements
- B. state / compression
- C. variance / pixels
- D. reward / latency

19. Two vehicles update their policies simultaneously. One vehicle's changing behavior alters the other's learning target. The effective environment becomes \_\_\_\_ and learning becomes \_\_\_\_.

A. non-stationary / coupled

B. observable / fixed

C. deterministic / independent

D. stationary / isolated

20. Four agents receive one delayed team reward. They cannot identify individual contributions. This creates a \_\_\_\_ assignment problem and makes credit \_\_\_\_ difficult.

A. reward / clipping

B. credit / assignment

C. state / transition

D. policy / sharing

21. Two agents must choose complementary actions to succeed. Independent learning may fail because success depends on joint \_\_\_\_ and coordinated \_\_\_\_.

A. states / rewards

B. actions / policies

C. episodes / observations

D. datasets / gradients

22. During training, a MARL system sees all agents' observations, but during deployment each agent sees only local data. This is centralized \_\_\_\_ with decentralized \_\_\_\_.

A. reward / learning

B. sampling / planning

C. execution / training

D. training / execution

23. In competitive MARL, one agent changes its policy and alters the strategies faced by others. This creates strategic \_\_\_\_ and reduces environmental \_\_\_\_.

A. sampling / accuracy

B. coupling / stationarity

C. reward / latency

D. compression / variance

24. A robot cannot observe a hidden obstacle directly and receives noisy sensor readings. The problem is partially \_\_\_\_, so the agent maintains a \_\_\_\_ state.

A. deterministic / policy

B. observable / belief

C. stationary / reward

D. episodic / gradient

25. A POMDP agent combines current evidence with previous observations before acting. Its decision uses a \_\_\_\_ state rather than only the current \_\_\_\_.

- A. reward / state
- B. gradient / episode
- C. policy / action
- D. belief / observation

26. The same sensor reading can lead to different actions because the observation history differs. Under partial observability, the agent therefore uses \_\_\_\_ information to infer the hidden \_\_\_\_.

- A. random / reward
- B. visual / gradient
- C. historical / state
- D. fixed / policy

27. An RL controller reduces waiting time by violating an unstated safety rule. The reward is \_\_\_\_, causing reward \_\_\_\_.

- A. normalized / clipping
- B. dense / sampling
- C. misspecified / hacking
- D. sparse / shaping

28. An RL system performs well in simulation but fails with real traffic. The problem reflects simulation-to-real \_\_\_\_ and requires stronger \_\_\_\_ testing.

- A. reward / rendering
- B. compression / encryption
- C. sampling / clipping
- D. shift / validation

29. A safety-critical system combines HRL, POMDP reasoning, and MARL. Hierarchy adds structural complexity, POMDP adds \_\_\_\_, and MARL adds agent \_\_\_\_.

- A. compression / latency
- B. normalization / variance
- C. sampling / encryption
- D. uncertainty / interaction